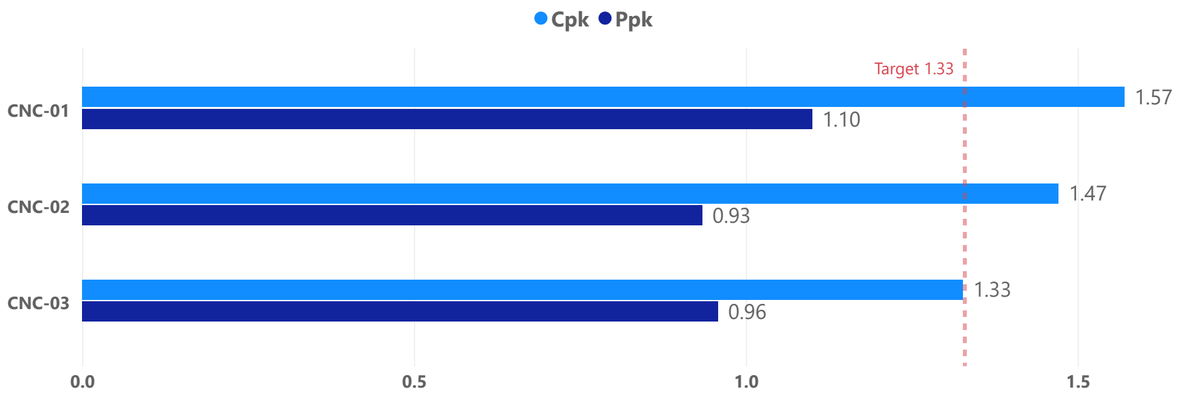
Brake Caliper Bore: Quality Overview

Jan–Jun 2026 · Bore Ø 42.000 ± 0.025 mm · Customer requirement: Ppk ≥ 1.33.

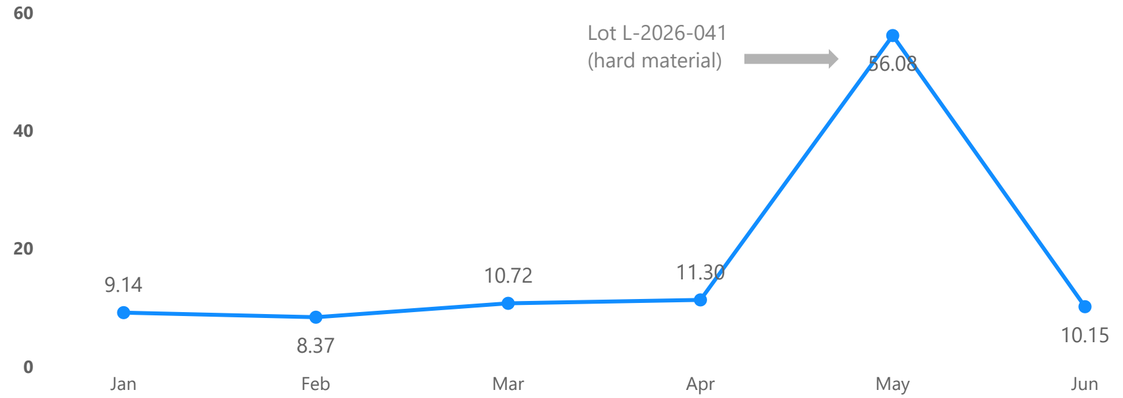
Shift: All | Machine: All | Month: All | Supplier: All

Parts produced	Defect rate per 1000	Rework cost	Scrap cost	Total defect cost	Machines at Ppk ≥ 1.33
253,374	17.72	€ 16.00K	€ 12.58K	€ 28.58K	0 of 3

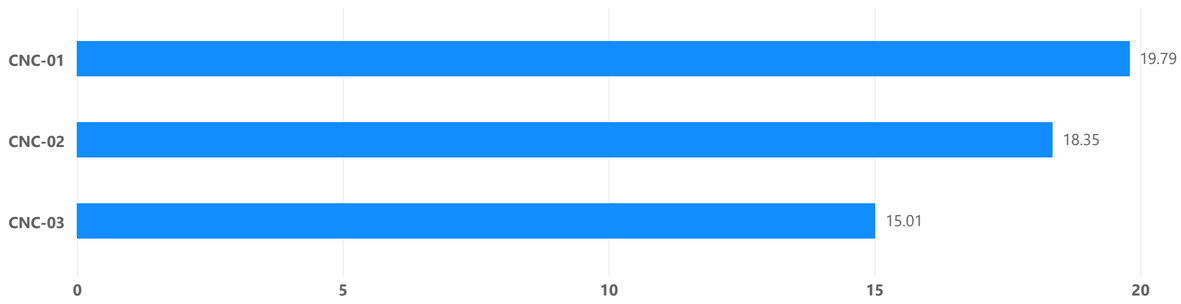
Process capability (target ≥ 1.33)



Defect rate by month (per 1,000 parts)



Defect rate by machine (per 1,000 parts)



Defect rate by shift (per 1,000 parts)



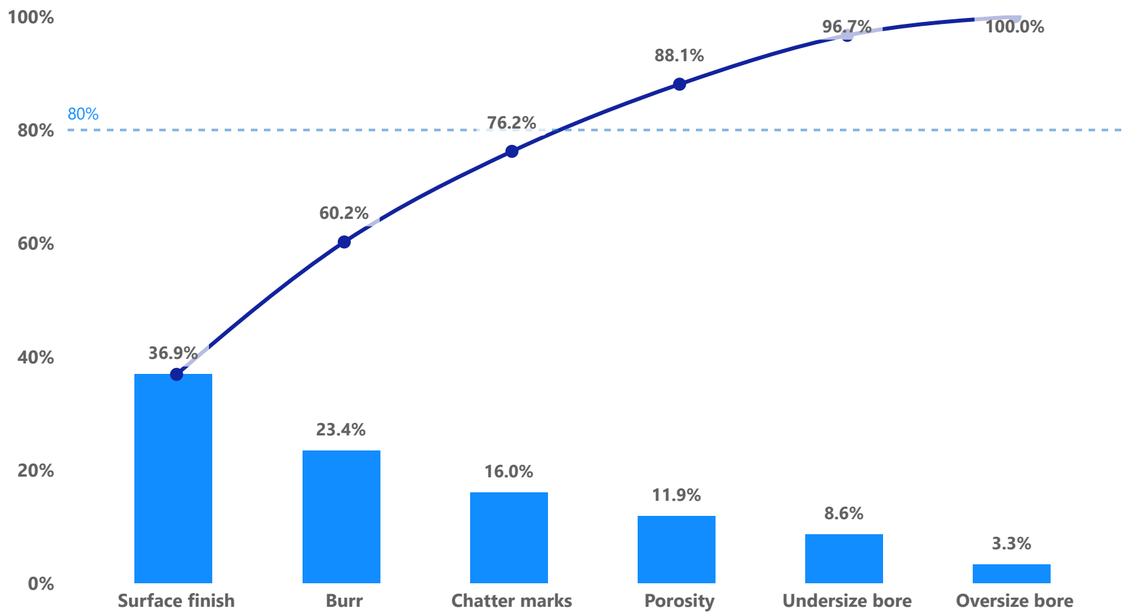
Defect Pareto

The two rankings disagree

Shift: All | Machine: All | Month: All | Supplier: All

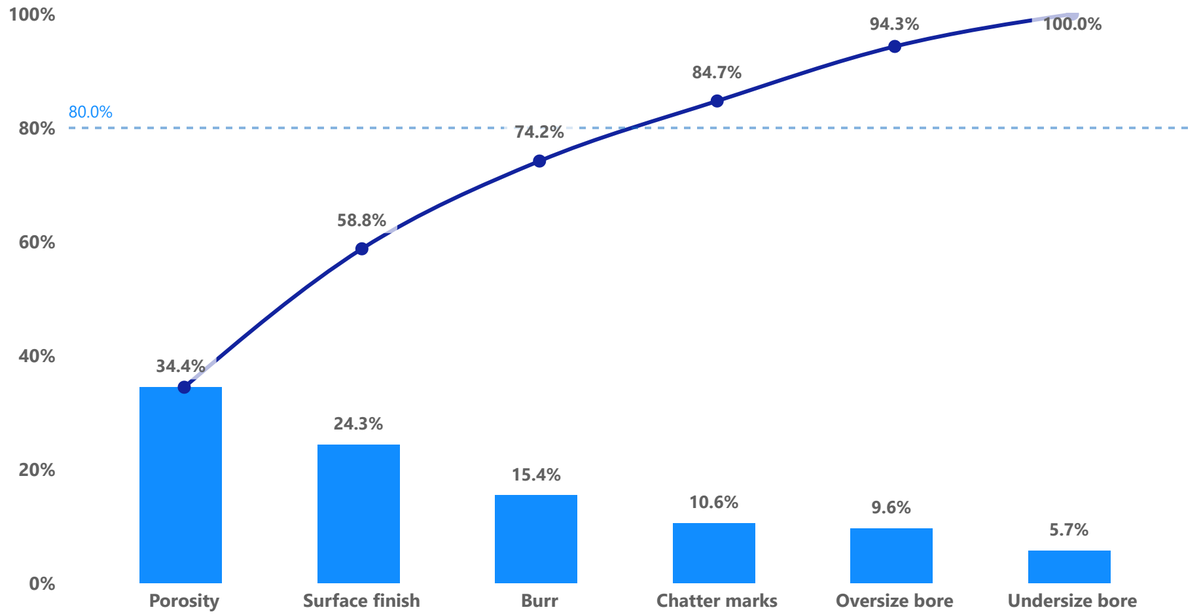
By count: Surface finish leads

4,489 defective parts · bars = share of total, line = cumulative share



By cost: porosity leads

€28,578 total defect cost · bars = share of total, line = cumulative share



Porosity is 4th by count but 1st by cost, because porosity parts are scrapped (€18.50 each) rather than reworked (€4.20). Prioritise by cost, not by count.

Statistical Process Control: X̄-R charts

One machine at a time · control limits from the January baseline · subgroups of 5 parts every 2 h

Machine

CNC-03

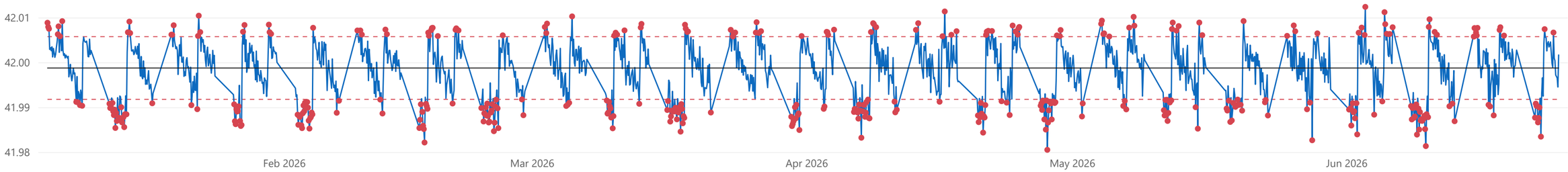
Date

1/5/2026 6/24/2026

LCL X-bar	UCL X-bar	Centre line	Subgroups shown	% any WE rule	% outside limits
41.9918	42.0057	41.9988	1441	69.0%	27.6%

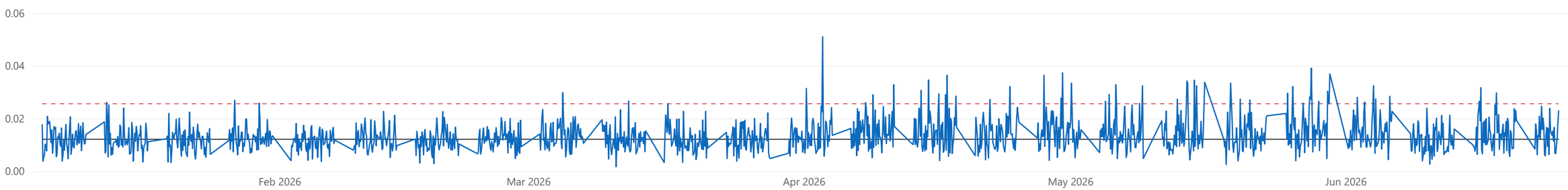
X̄ chart: subgroup mean (limits from January)

blue = subgroup mean · red dashed = control limits · grey = centre line



R chart: subgroup range

blue = subgroup range · red dashed = upper control limit · grey = average range (R̄)



How to read: the sawtooth is tool wear (jump up at each tool change, drift down as the tool wears). Try: CNC-02, 9 Mar–3 Apr (offset error) · CNC-01, 4–22 May (tool breakage on 12 May) · CNC-03, R chart from April (night-shift spread).